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Introduction

This is a sample of the work done for evaluating foundational models in a pathology pipeline using several slide samples and their ground truths.

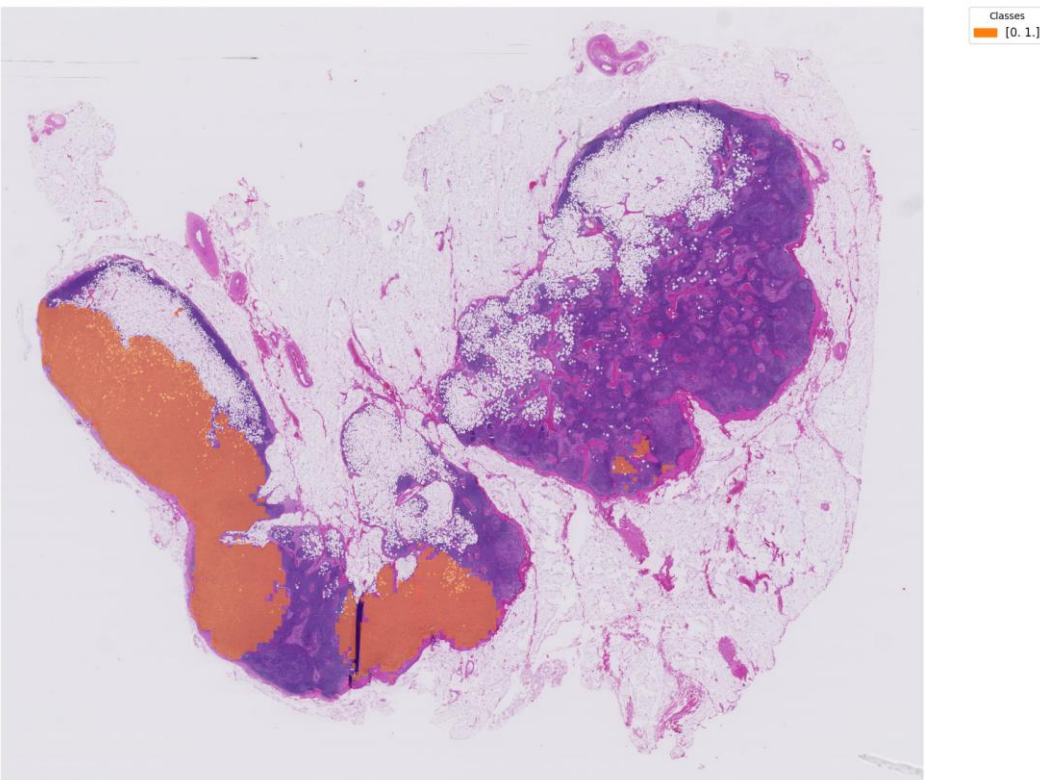
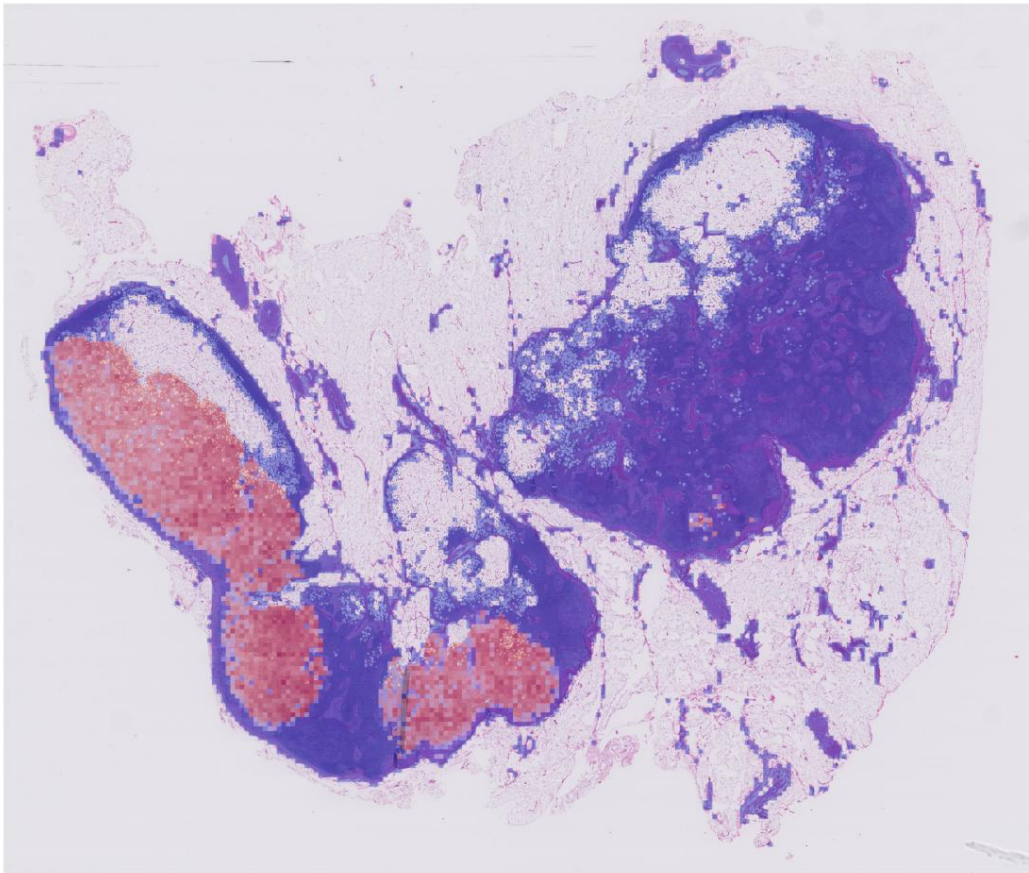
- The model operates multimodal, including language, annotations, slide labels etc.
- The samples show annotations and slide label capabilities here (as preview)
- The samples below are strictly under test set which had not been trained or evaluated by the model.
- The model results are hand-picked so that it better illustrates the capabilities and weaknesses of the model.
- The results here shown have all predicted the **slide labels** correctly, along with the model's reasoning (attention map)
- Attention map means: what the model looks at to give its prediction

Task details:

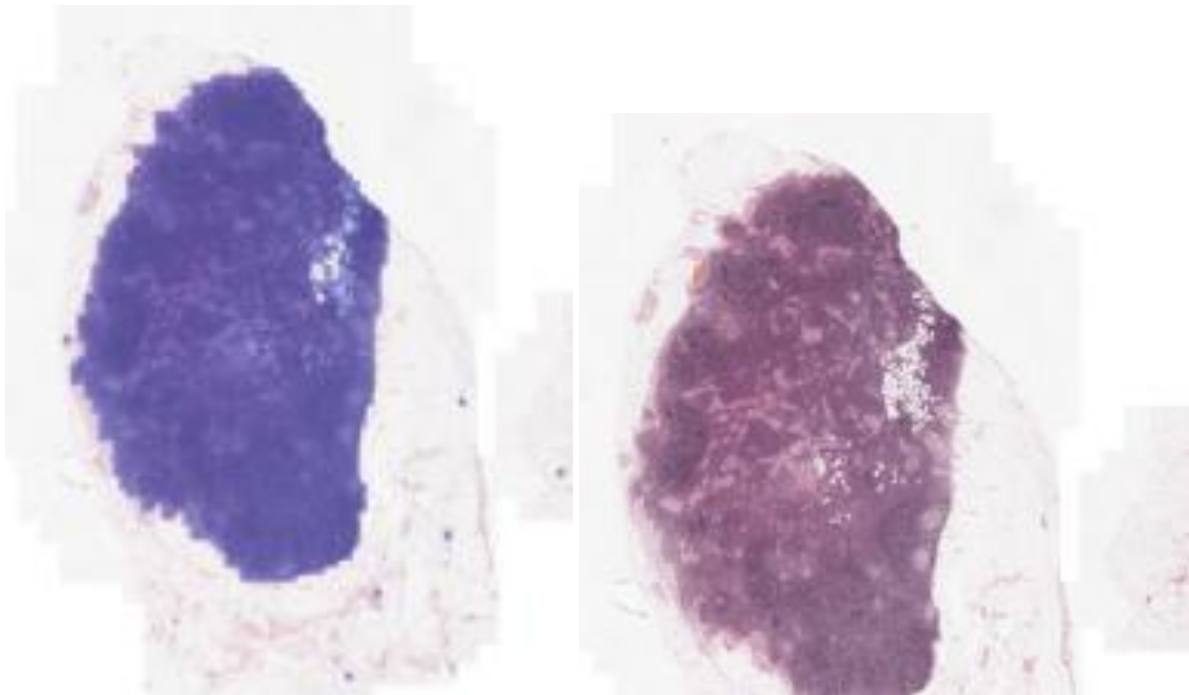
1. Breast metastasis (400 total samples)
 - a. Negative
 - b. Micro
 - c. macro
2. Gleason scoring (10,000 total samples)
 - a. Stroma
 - b. Normal
 - c. Gleason 3
 - d. Gleason 4
 - e. Gleason 5

Breast metastasis

breast metastasis macro (top is prediction, bottom is test set)



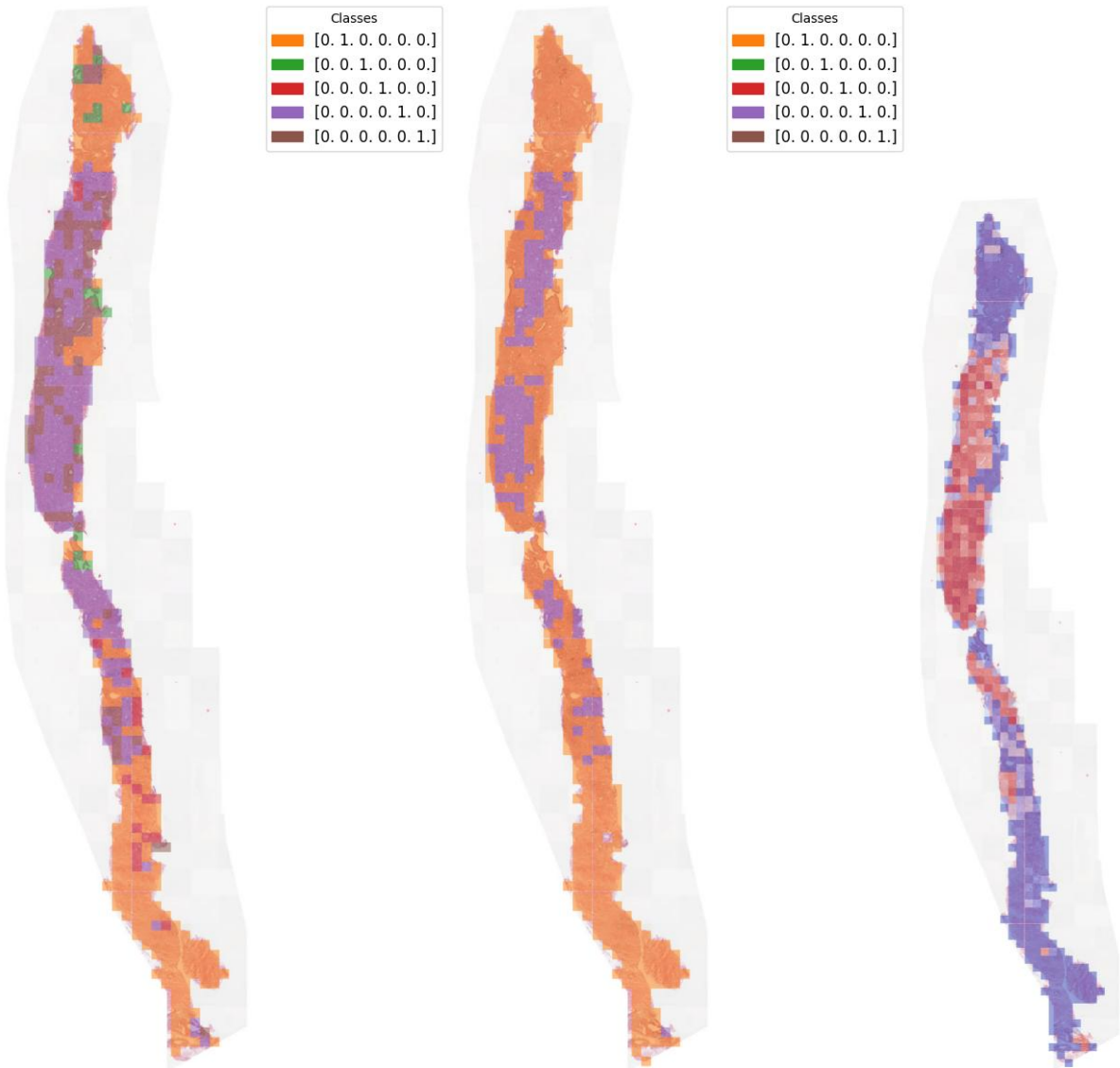
breast metastasis micro (left is prediction and right is test set)



Gleason Scoring

Gleason score of 4

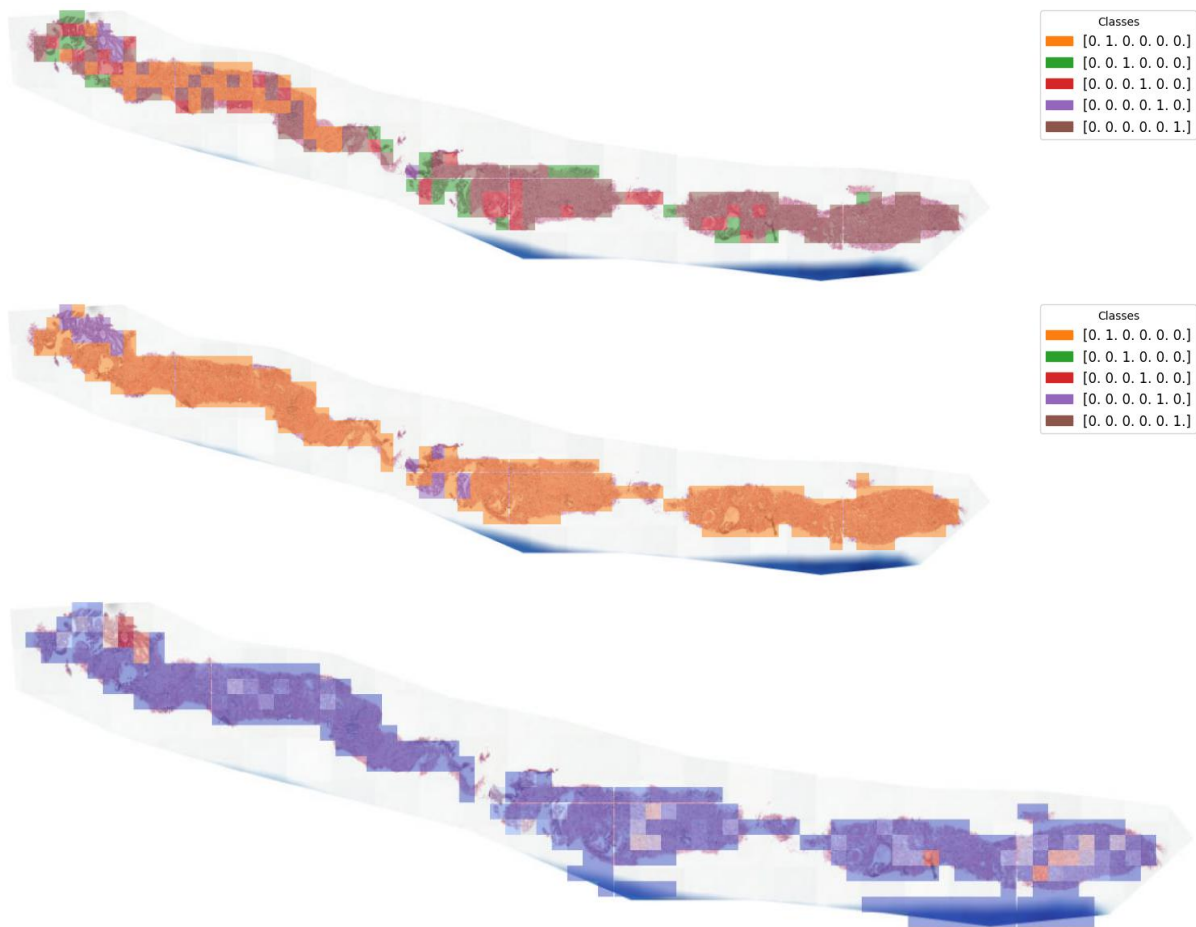
- left is prediction
- mid is test set
- right is attention map



Gleason score of 4+3 (left is prediction, right is test set, bottom is attention map)

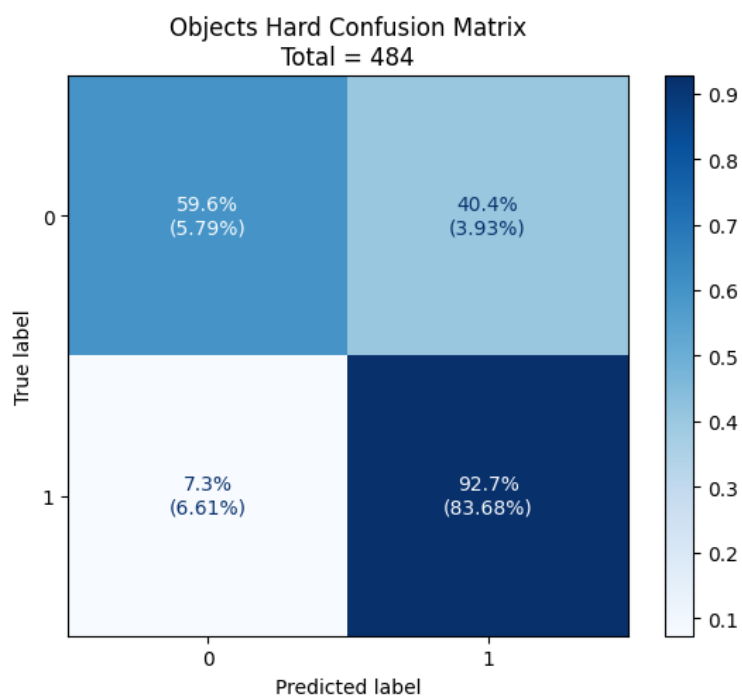
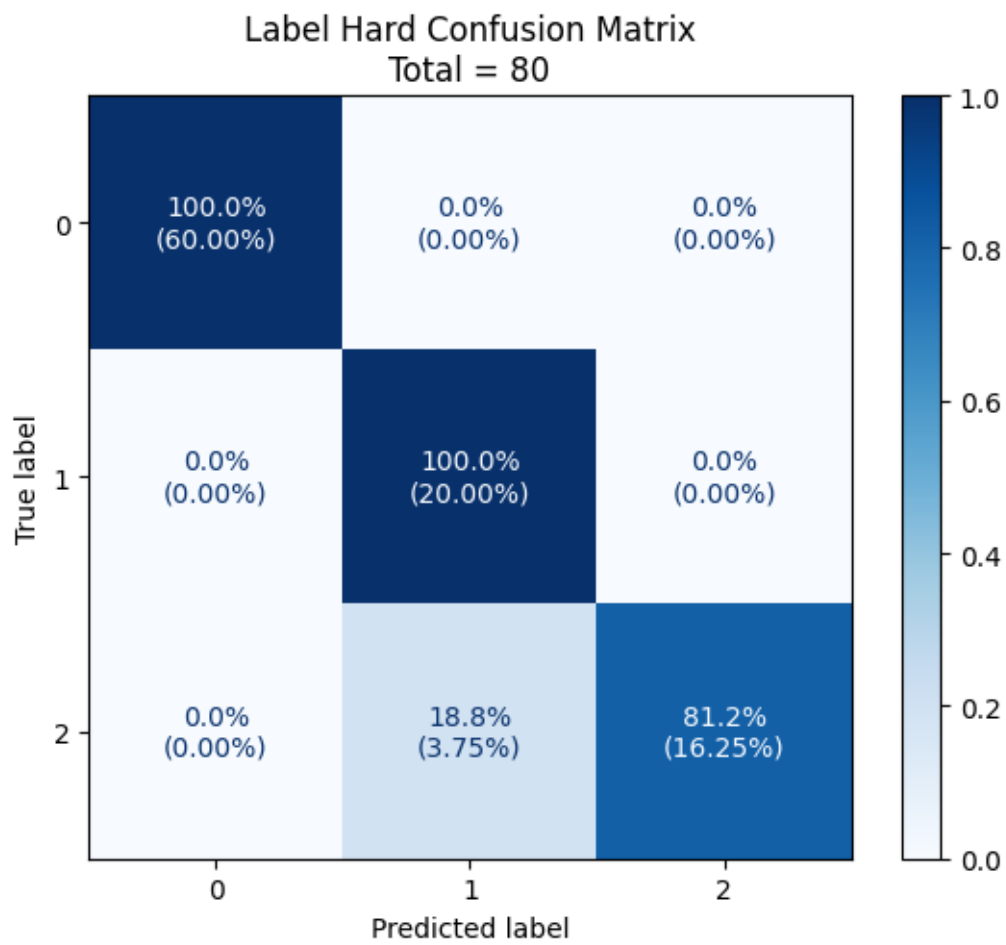


Gleason Score of 5+4 (top is prediction, mid is test set, bottom is attention map)



Metrics

Breast metastasis



Gleason Scoring

